QQKRLS

Quantile-on-Quantile Kernel-Based Regularized Least Squares

A Complete Technical Guide

From Zero to Expert — For Researchers & Practitioners

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PyPI: https://pypi.org/project/qqkrls/
GitHub: https://github.com/merwanroudane/qqkrls

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1 Introduction

Who Is This Guide For?

This guide is written for **researchers, graduate students, and practitioners** in economics, finance, and environmental science who want to apply advanced nonparametric methods without requiring deep machine-learning expertise. We start from basic regression concepts and build up to the full QQKRLS estimator.

1.1 Motivation: Why Go Beyond OLS?

Ordinary Least Squares (OLS) is the workhorse of empirical economics. Given a model $y = X\beta + \varepsilon$, OLS provides the best linear unbiased estimator (BLUE) under the Gauss–Markov assumptions. However, OLS imposes strong restrictions:

1. **Linearity**: The conditional expectation $\mathbb{E}[y \mid x]$ must be linear in x.
2. **Additivity**: Each predictor's effect is separable — no automatic interactions.
3. **Homogeneity**: The effect β_j is *constant* across the entire distribution.

In reality, economic relationships are often **nonlinear**, **interactive**, and **heterogeneous across the distribution**. For example, the effect of energy consumption on environmental quality may differ at low vs. high consumption levels.

The Problem with Misspecification

If the true data-generating process is nonlinear but we fit a linear model, OLS estimates are **biased and inconsistent**. Diagnostic tests (Ramsey RESET, BDS) can detect this misspecification, but the solution requires a more flexible estimator.

1.2 The QQKRLS Framework

The `qqkrls` library provides a unified solution through four methods:

Method	Description
KRLS	Kernel Regularized Least Squares — nonparametric regression in a reproducing kernel Hilbert space
QQ Regression	Quantile-on-Quantile analysis — examines how effects vary across both distributions
WQR	Weighted Quantile Regression — kernel-weighted quantile estimates
QQKRLS	The combined estimator — nonparametric marginal effects across all quantile pairs

2 Kernel Regularized Least Squares (KRLS)

2.1 Intuition: From Linear to Nonparametric

Consider the classical regression problem: given data $\{(x_i, y_i)\}_{i=1}^N$ where $x_i \in \mathbb{R}^D$ and $y_i \in \mathbb{R}$, we want to estimate the conditional expectation function $f(x) = \mathbb{E}[y \mid x]$.

- **OLS**: Restricts $f(x) = x^\top \beta$ — a linear function.
- **KRLS**: Allows f to be *any smooth function* in a reproducing kernel Hilbert space $\mathcal{H}_K$.

The key insight is the **kernel trick**: instead of explicitly specifying a functional form, KRLS defines similarity between observations using a *kernel function* and learns a flexible, data-driven relationship.

2.2 The Gaussian Kernel

Definition: Gaussian (RBF) Kernel

The Gaussian kernel measures the similarity between two observations x_i and x_j:

$$K(x_i, x_j) = \exp\left(-\frac{\|x_i - x_j\|^2}{\sigma}\right) \tag{1}$$

where $\sigma > 0$ is the **bandwidth** parameter controlling the smoothness of the function.

Properties:

- $K(x_i, x_i) = 1$ — each observation is perfectly similar to itself.
- $K(x_i, x_j) \to 0$ as $\|x_i - x_j\| \to \infty$ — distant observations have negligible influence.
- **Large** σ: Broader influence $\to$ smoother function (closer to linear).
- **Small** σ: Narrower influence $\to$ more flexible (risk of overfitting).

Default Bandwidth

Following Hainmueller & Hazlett (2014), KRLS sets $\sigma = D$ (the number of predictors). This heuristic works well in practice and avoids expensive bandwidth search.

2.3 The KRLS Optimization Problem

KRLS Objective Function

KRLS solves the following Tikhonov-regularized optimization:

$$\hat{f} = \arg\min_{f \in \mathcal{H}_K} \underbrace{\sum_{i=1}^{N} (y_i - f(x_i))^2}_{\text{Data fidelity}} + \underbrace{\lambda \|f\|^2_{\mathcal{H}_K}}_{\text{Regularization}} \tag{2}$$

where $\lambda > 0$ controls the trade-off between fitting the data and keeping the function smooth.

Interpretation:

- The first term minimizes the sum of squared residuals (like OLS).
- The second term penalizes complexity in the RKHS norm, preventing overfitting.
- $\lambda \to 0$: Interpolates the data (overfitting).
- $\lambda \to \infty$: Converges to the sample mean (underfitting).

2.4 The Representer Theorem and Closed-Form Solution

Representer Theorem (Scholkopf & Smola, 2002)

The minimizer of Eq. (2) has the finite representation:

$$\hat{f}(x) = \sum_{i=1}^{N} c_i \cdot K(x, x_i) \tag{3}$$

where $\mathbf{c} = (c_1, \ldots, c_N)^\top$ is a vector of N coefficients.

This is remarkable: although the RKHS is infinite-dimensional, the optimal solution only depends on N parameters. The predicted value at any point x is a **weighted sum of kernel similarities** to all training observations.

2.4.1 Eigendecomposition Solution

Substituting Eq. (3) into Eq. (2) and taking the first-order condition:

$$\mathbf{c}^* = (\mathbf{K} + \lambda \mathbf{I})^{-1} \mathbf{y} \tag{4}$$

where $\mathbf{K}$ is the $N \times N$ kernel matrix with $K_{ij} = K(x_i, x_j)$.

For numerical stability, KRLS uses the eigendecomposition $\mathbf{K} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^\top$:

$$\boxed{\mathbf{c}^* = \mathbf{V}(\mathbf{\Lambda} + \lambda \mathbf{I})^{-1} \mathbf{V}^\top \mathbf{y}} \tag{5}$$

Computational Advantage

The eigendecomposition is computed **once** and reused for different values of λ, making leave-one-out cross-validation extremely efficient.

2.5 Regularization Parameter Selection via LOO-CV

Leave-One-Out Cross-Validation

The LOO-CV error is:

$$\mathrm{LOO}(\lambda) = \sum_{i=1}^{N} \left(\frac{y_i - \hat{f}_\lambda(x_i)}{1 - H_{ii}(\lambda)} \right)^2 \tag{6}$$

where $H_{ii}(\lambda)$ is the i-th diagonal of the hat matrix $= \mathbf{K}(\mathbf{K} + \lambda \mathbf{I})^{-1}$.

KRLS selects λ^* by minimizing this LOO error, which can be computed **without actually refitting** the model N times (using the eigendecomposition).

2.6 Pointwise Marginal Effects

Unlike OLS (which gives one constant coefficient per variable), KRLS provides a **marginal effect for each observation**. This is computed analytically:

KRLS Pointwise Marginal Effect

The marginal effect of predictor d at observation j is:

$$\frac{\partial \hat{f}(x_j)}{\partial x_j^{(d)}} = \frac{-2}{\sigma} \sum_{i=1}^{N} c_i \cdot K(x_j, x_i) \cdot \left(x_i^{(d)} - x_j^{(d)} \right) \tag{7}$$

Key insight: The marginal effect varies by observation. If these pointwise effects show substantial variation (different at Q25 vs. Q75), the relationship is genuinely nonlinear, justifying KRLS over OLS.

Interpreting Marginal Effects

- **Average Marginal Effect**: $\overline{\mathrm{ME}}_d = \frac{1}{N} \sum_{j=1}^{N} \frac{\partial \hat{f}}{\partial x_j^{(d)}}$ — comparable to OLS $\hat{\beta}_d$.
- **Heterogeneity**: If Q25 $\neq$ Q75 of the effects, the relationship is nonlinear.
- **Plots**: Scatter plots of marginal effects vs. predictor values reveal the functional form.

3 Quantile-on-Quantile (QQ) Regression

3.1 Motivation: Beyond the Mean

Standard regression estimates the **conditional mean** $\mathbb{E}[y \mid x]$. But economic relationships often differ across the distribution:

- Does energy consumption affect environmental quality *differently* when the footprint is already high vs. low?
- Is the GDP–pollution nexus stronger during economic booms (high GDP quantiles) or recessions (low GDP quantiles)?

Quantile regression (Koenker, 2005) addresses the first question by estimating conditional quantiles $Q_\tau(y \mid x)$ for different $\tau \in (0, 1)$. But it still assumes a *fixed* effect of x regardless of where x falls in its own distribution.

Quantile-on-Quantile Regression (Sim & Zhou, 2015)

QQ regression creates a **two-dimensional mapping** $\beta(\theta, \tau)$ that captures how the θ-th quantile of X affects the τ-th conditional quantile of Y. This reveals distributional heterogeneity in *both* variables simultaneously.

3.2 The QQ Regression Procedure

Algorithm 1 Quantile-on-Quantile Regression

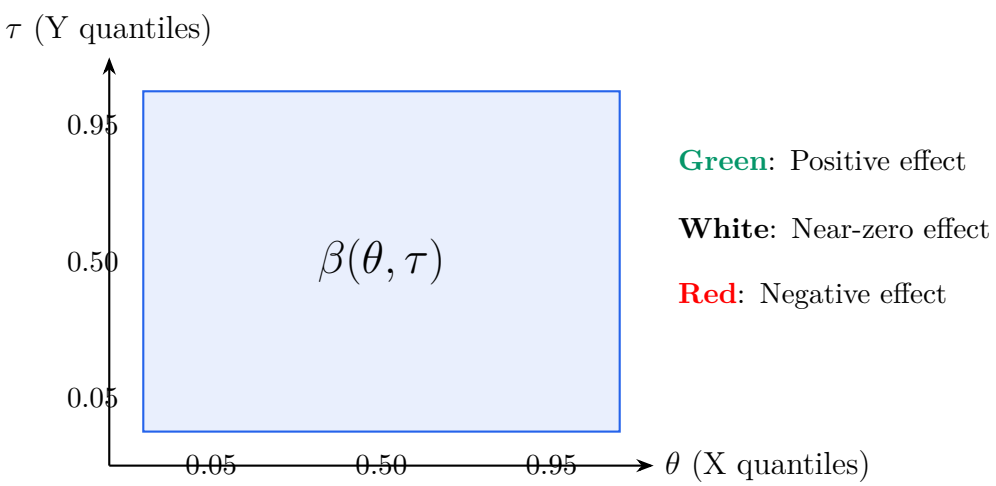
Require: Data $\{(x_i, y_i)\}_{i=1}^N$, quantile grids $\Theta = \{\theta_1, \ldots, \theta_J\}$ and $\mathcal{T} = \{\tau_1, \ldots, \tau_K\}$
Ensure: Coefficient matrix $\beta(\theta, \tau)$ of dimension $K \times J$

```
for each θ ∈ Θ do
    Compute threshold: q_θ = Q_X(θ)                    ▷ θ-th quantile of X
    Subset: S_θ = {(x_i, y_i) : x_i ≤ q_θ}
    for each τ ∈ T do
        Estimate τ-th quantile regression of Y on X using S_θ
        Store β̂(θ, τ)
    end for
end for
return Matrix [β̂(θ_j, τ_k)]_{k,j}
```

3.3 Interpreting the QQ Coefficient Matrix

The result is a $K \times J$ matrix where:

- **Rows** (τ): Quantiles of Y (the dependent variable)
- **Columns** (θ): Quantiles of X (the independent variable)
- **Cell** $\beta(\theta, \tau)$: The marginal effect of X on Y at that specific distributional location



4 The QQKRLS Estimator

4.1 Combining KRLS with QQ Regression

QQKRLS (Adebayo et al., 2024) replaces the linear quantile regression step in QQ with **KRLS**, creating a fully nonparametric estimator:

The QQKRLS Algorithm

For each X-quantile θ and Y-quantile τ:

1. **Subset**: Select observations where $x_i \leq Q_X(\theta)$
2. **Fit KRLS**: Apply kernel regression on the subset $\rightarrow$ obtain pointwise marginal effects $\left\{\frac{\partial \hat{f}_{[\theta]}}{\partial x}\right\}$
3. **Extract quantile**: The coefficient is the τ-th quantile of the marginal effects:

$$\boxed{\beta(\theta,\tau) = Q_\tau\left(\left\{\frac{\partial \hat{f}_{[\theta]}(x_j)}{\partial x}\right\}_{j\in\mathcal{S}_\theta}\right)} \tag{8}$$

4.2 Detailed QQKRLS Algorithm

Algorithm 2 QQKRLS Estimation with Bootstrap Inference

Require: $\mathbf{y}, \mathbf{x}$, quantile grids $\Theta, \mathcal{T}$, bootstrap replications B, min. obs. $n_{\min}$
Ensure: $\hat{\beta}(\theta,\tau)$, $\mathrm{SE}(\theta,\tau)$, $t(\theta,\tau)$, $p(\theta,\tau)$

```
for each θ ∈ Θ do
    q_θ ← Q_X(θ);    S_θ ← {i : x_i ≤ q_θ}
    if |S_θ| ≥ n_min then
        Fit KRLS on (x_{S_θ}, y_{S_θ}) with derivative=True
        Extract pointwise marginal effects {d_j}_{j∈S_θ}
        for each τ ∈ T do
            β̂(θ,τ) ← Q_τ({d_j})                          ▷ τ-th quantile of effects
            for b = 1, ..., B do                          ▷ Bootstrap for SE
                Draw bootstrap sample with replacement from S_θ
                Fit KRLS on bootstrap sample
                β̂^(b)(θ,τ) ← Q_τ of bootstrap marginal effects
            end for
            ŜE ← std(β̂^(1), ..., β̂^(B))
            t ← β̂/ŜE;    p ← 2Φ(−|t|)
        end for
    end if
end for
```

4.3 Why QQKRLS Improves Over Standard QQ

	Standard QQ	QQKRLS
Inner estimator	Linear quantile regression	KRLS (nonparametric)
Functional form	Assumes linearity within subsets	No functional form assumption
Interactions	Not captured	Automatic via Gaussian kernel
Marginal effects	Constant within each subset	Pointwise (varies by obs.)
Inference	Asymptotic	Bootstrap (finite-sample valid)

4.4 Statistical Inference

4.4.1 Bootstrap Standard Errors

For each cell (θ, τ), QQKRLS uses the **paired bootstrap**:

1. Draw B bootstrap samples (with replacement) from the subset $\mathcal{S}_\theta$
2. Re-estimate KRLS and extract the τ-th quantile of marginal effects
3. The bootstrap standard error: $\widehat{\mathrm{SE}}(\theta, \tau) = \mathrm{std}\{\hat{\beta}^{(1)}, \ldots, \hat{\beta}^{(B)}\}$

4.4.2 Test Statistics and P-values

Under the null $H_0 : \beta(\theta, \tau) = 0$:

$$t(\theta, \tau) = \frac{\hat{\beta}(\theta, \tau)}{\widehat{\mathrm{SE}}(\theta, \tau)} \xrightarrow{d} \mathcal{N}(0, 1) \tag{9}$$

$$p(\theta, \tau) = 2\Phi(-|t(\theta, \tau)|) \tag{10}$$

Significance Conventions

- ***: $p < 0.01$ (highly significant)
- **: $p < 0.05$ (significant at 5%)
- *: $p < 0.10$ (marginally significant)

These stars are displayed on the heatmap to show which cells have statistically significant effects.

5 Pre-Estimation Diagnostic Tests

Before applying KRLS/QQKRLS, researchers must **justify** the use of nonparametric methods. The `qqkrls` library includes a complete diagnostic battery.

5.1 Ramsey RESET Test

Ramsey RESET (Regression Equation Specification Error Test)

Tests whether the linear model is correctly specified by including powers of fitted values:

$$y = X\beta + \gamma_2 \hat{y}^2 + \gamma_3 \hat{y}^3 + \varepsilon \tag{11}$$

- H_0: Linear model is correctly specified ($\gamma_2 = \gamma_3 = 0$)
- H_1: Nonlinear terms are significant $\rightarrow$ **use KRLS**

5.2 Breusch–Pagan Test

Breusch–Pagan Test for Heteroskedasticity

Tests whether the variance of residuals depends on the predictors:

$$\hat{\varepsilon}^2 = X\alpha + u \tag{12}$$

- H_0: Homoskedasticity ($\mathrm{Var}[\varepsilon \mid X] = \sigma^2$)
- H_1: Heteroskedasticity $\rightarrow$ **effects differ across distribution** $\rightarrow$ use quantile-based analysis

5.3 BDS Test

BDS Test (Brock, Dechert, Scheinkman)

A nonparametric test for serial independence and nonlinear structure in the residuals:

- H_0: Residuals are i.i.d.
- H_1: Nonlinear dependence exists $\rightarrow$ **justifies KRLS/QQKRLS**

The test uses the correlation integral at embedding dimension m:

$$C_m(\epsilon) = \lim_{N\to\infty} \frac{2}{N(N-1)} \sum_{i<j} \mathbf{1}\left[\|x_i^m - x_j^m\| < \epsilon\right] \tag{13}$$

Under the null, $C_m(\epsilon) = [C_1(\epsilon)]^m$. The BDS statistic tests this equality.

5.4 Jarque–Bera Normality Test

Jarque–Bera Test

Tests whether the data follows a normal distribution:

$$JB = \frac{N}{6}\left(S^2 + \frac{(K-3)^2}{4}\right) \sim \chi^2(2) \tag{14}$$

where S = skewness, K = kurtosis. Rejection suggests **non-Gaussian distributions**, motivating quantile-based methods.

5.5 Andrews Parameter Stability Test

Andrews Sup-F Test

Tests for structural breaks in the relationship by computing F-statistics at all possible break points:

- **Max-F**: Supreme of F-statistics (detects single break)
- **Exp-F**: Exponential average (robust to multiple breaks)
- **Ave-F**: Simple average (overall instability)

Parameter instability justifies quantile-based methods that allow coefficients to vary.

6 Using the qqkrls Python Library

6.1 Installation

```
pip install qqkrls
```

6.2 Complete Workflow

```
import numpy as np
import pandas as pd
from qqkrls import (
    krls, qqkrls, predict_krls,
    plot_qqkrls_heatmap, plot_qqkrls_3d,
    plot_qqkrls_contour, plot_qqkrls_pvalue,
    plot_krls_derivatives, plot_krls_fit,
    descriptive_statistics, linearity_test_battery,
    bds_test, jarque_bera, print_diagnostics,
)

# Step 1: Load data
y = df['EF'].values
```

```
X = df[['Energy', 'CO2', 'GDP']].values

# Step 2: Pre-estimation diagnostics
diag = linearity_test_battery(y, X, col_names=['Energy','CO2','GDP'])
print_diagnostics(diag)

# Step 3: Fit KRLS
fit = krls(X, y, derivative=True, vcov=True, verbose=1)

# Step 4: QQKRLS estimation
taus = np.arange(0.05, 1.0, 0.05)
result = qqkrls(y=y, x=df['Energy'].values,
                y_quantiles=taus, x_quantiles=taus,
                n_boot=200, verbose=True)

# Step 5: Visualize
plot_qqkrls_heatmap(result, colorscale='paper',
                    show_stars=True, save_path='heatmap.png')
plot_qqkrls_3d(result, save_path='surface.png')
```

6.3 Key API Functions

Function	Description
`krls(X, y)`	Fit KRLS model with LOO cross-validation
`qqkrls(y, x)`	Run QQKRLS with bootstrap inference
`predict_krls(fit, X_new)`	Out-of-sample prediction
`plot_qqkrls_heatmap()`	Paper-style coefficient heatmap
`plot_qqkrls_3d()`	MATLAB-style 3D surface plot
`plot_qqkrls_contour()`	Filled contour plot
`plot_qqkrls_pvalue()`	P-value significance map
`plot_qqkrls_panel()`	Multi-variable panel
`plot_krls_derivatives()`	Marginal effect scatter + density
`plot_krls_fit()`	Fitted vs actual with R^2
`descriptive_statistics()`	LaTeX descriptive statistics table
`qqkrls_coefficient_table()`	LaTeX coefficient matrix
`krls_summary_table()`	LaTeX KRLS summary
`export_results_csv()`	CSV export of all results
`linearity_test_battery()`	RESET + BP + JB + BDS
`bds_test()`	BDS independence test
`parameter_stability_test()`	Andrews sup-F, exp-F, ave-F
`jarque_bera()`	Jarque–Bera normality test

7 Interpreting QQKRLS Results

7.1 Reading the Heatmap

The QQKRLS heatmap is the primary output. Each cell (θ, τ) represents one nonparametric marginal effect:

Common Misinterpretations to Avoid

- The axes are **quantiles**, not raw values of the variables.
- $\theta = 0.05$ means "when X is at its 5th percentile," not "when $X = 0.05$."
- A positive coefficient means: at that distributional location, increasing X *increases* Y.

7.2 Pattern Recognition

Pattern	Economic Interpretation
Uniform positive	X always increases Y (homogeneous positive effect)
Gradient left-to-right	Effect strengthens as X increases (nonlinear amplification)
Gradient bottom-to-top	Effect stronger when Y is high (asymmetric vulnerability)
Diagonal pattern	Effect strongest at matching quantiles (distributional symmetry)
Quadrant split	Effect changes sign across the distribution (regime switching)
Significance clusters	Only certain quantile pairs show significant effects

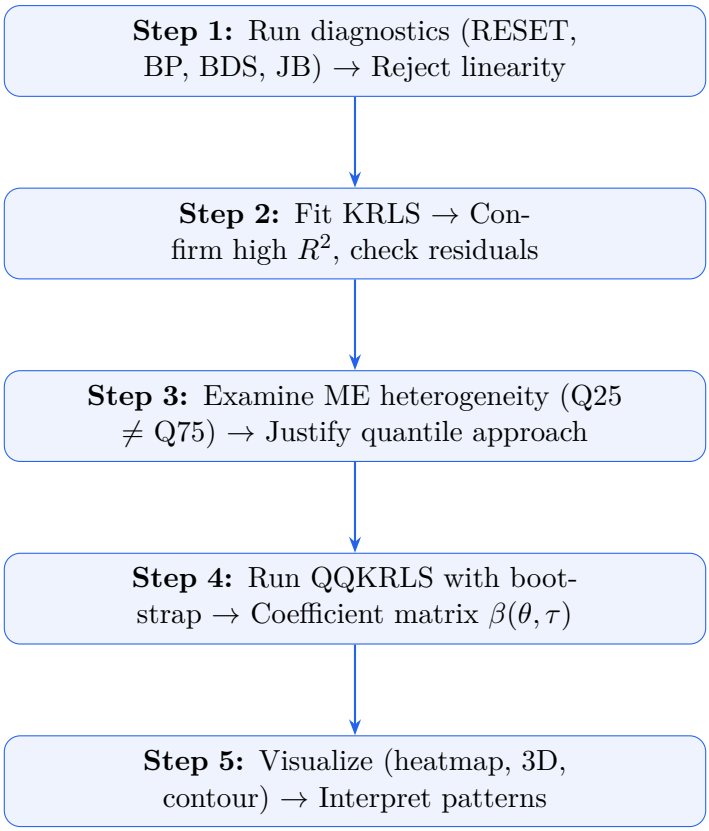
7.3 Writing Up Results for a Paper

Template for Results Section

"Table X presents the QQKRLS estimates of the marginal effect of [Variable] on [Dependent Variable] across 19 quantile pairs ($\theta, \tau \in \{0.05, 0.10, \ldots, 0.95\}$). The pre-estimation diagnostics (Ramsey RESET: $F = \ldots$, $p = \ldots$; BDS: $z = \ldots$, $p < 0.01$) reject linearity and i.i.d. residuals, justifying the nonparametric QQKRLS approach. The heatmap in Figure Y reveals substantial heterogeneity across the distribution. At lower quantiles of [Variable] ($\theta < 0.30$), the effect on [DV] is [negative/insignificant], while at upper quantiles ($\theta > 0.70$), the effect becomes [significantly positive]. This asymmetry suggests that [economic interpretation].
Of the 361 quantile-pair cells, [N] ([%]%) are significant at the 5% level, concentrated in the [describe region] of the heatmap. These findings are robust to the bootstrap inference with $B = 200$ replications."

7.4 Methodological Justification Workflow

The recommended workflow for any QQKRLS paper follows five steps:


Step 1: Run diagnostics (RESET, BP, BDS, JB) → Reject linearity
Step 2: Fit KRLS → Confirm high R^2, check residuals
Step 3: Examine ME heterogeneity (Q25 ≠ Q75) → Justify quantile approach
Step 4: Run QQKRLS with bootstrap → Coefficient matrix $\beta(\theta, \tau)$
Step 5: Visualize (heatmap, 3D, contour) → Interpret patterns

8 References

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